

Manas Kalangan

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EDUCATION

University at Buffalo, The State University of New York

Buffalo, USA

Master of Science in Computer Science and Engineering, GPA: 3.55/4

Aug 2024 - May 2026

WORK EXPERIENCE

Rocket Mortgage (Default Loan Servicing)

Software Engineer Intern

May 2025 - Aug 2025

- Built high-scale .NET 8 REST APIs improving mortgage approval workflows, reducing escalations by 30%
- Upgraded AWS Lambda functions with Terraform, ensuring fault-tolerant deployments and certificate renewals
- Used SQL Server to design schemas, write queries, and optimize data operations aligned with compliance requirements.
- Applied AI-powered tools (GitHub Copilot) to accelerate unit testing and debugging, boosting code quality

Amdocs Ltd

Software Engineer

Jan 2024 - Jul 2024

- Collaborated within a Agile/Scrum team of 6 to enhance Ginger - an open source testing tool, using C# and WPF
- Optimized the web automation testing feature, reducing the test suite execution time by 60-90%
- Refactored legacy codebases applying SOLID and OOP design principles, improving developer experience
- Redesigned the OCR feature to enhance PDF readability, leading to a 40% improvement in text clarity
- Authored ~200 unit tests using MSTest framework for new and legacy features, achieving 90% code coverage
- Migrated LiteDB from version 4.1 to 5.0, enhancing data storage and retrieval efficiency while preventing file corruption

Amdocs Ltd

Associate Software Engineer

Jul 2022 - Dec 2023

- Built internal support web app using ASP.NET MVC, improving client collaboration and issue tracking
- Resolved around 3,000 defects and 300 critical bugs, significantly boosting software stability and user satisfaction
- Boosted test coverage by 60% using Selenium for advanced web component testing, enhancing UI stability and reliability

UNIVERSITY PROJECTS

LegitReach - [demo](#)

Feb 2026 - Present

Personalized newsletter for online sellers (NextJs, PostgreSQL(Supabase), Vercel)

- Built an AI-powered lead curation engine in a team of 2, by automating Reddit, reducing manual research time by 80%
- Built a secure credit-based system with Stripe, automating user quota management and real-time payment verification
- Developed a real-time push-based curation system with Redis, enabling background processing of CPU-intensive tasks
- Scaled to 3 paying customers, rapidly iterating on AI features through direct user feedback loops.

ArtAtlas - [demo](#)

Dec 2025 - Jan 2025

Explainable Hybrid Art Search Engine (Python, PostgreSQL, pgvector, React)

- Built an end-to-end, explainable art search product that retrieves artworks and clearly shows why each result appears.
- Developed a hybrid search system (keyword + semantic) with a deterministic concept layer to avoid black-box behavior.
- Designed an evidence-backed explanation that attaches human-readable reasoning without impacting performance.

News Feed Service - [demo](#)

Nov 2025 - Dec 2025

Twitter-like Timeline Service (NodeJs, PostgreSQL, Redis, Kafka)

- Designed and implemented timeline service, modeling with a hybrid fan-out strategy to support high-follower users.
- Implemented an asynchronous write pipeline that persists posts and offloads fan-out processing to Kafka consumers.
- Developed a read-optimized feed service that merges Redis inboxes with the pulled posts from high-follower users.

Progression Palette - [demo](#)

Mar 2025 - Apr 2025

Emotion-Aware Music Generation System (Python, music21, Wikifonia)

- Built an interpretable music generation system that creates chord progressions based on emotional intent and theory.
- Avoided black-box ML by using rule-based harmony and probabilistic sampling for controllable, varied outputs.
- Designed an emotion-driven harmonic decoration layer that adjusts chord extensions in a musically explainable way.

SKILLS

Programming: C#, JavaScript, TypeScript, Python

Libraries/Frameworks: .NET Framework, ReactJS, Node.js, Express.js, WPF, MSTest

Cloud & DevOps Tools: AWS Lambda, AWS RDS, Terraform, Azure DevOps, Git, GitHub

Development Tools & IDEs: Visual Studio, VS Code, Claude Code, GitHub Copilot

Database & Caching Systems: MongoDB, MS SQL, MySQL, PostgreSQL, Redis